

# The VLM→Action Read-Out Interface

Phase 1: what governs a frozen-VLM manipulation policy, and what does not

|            |                                                          |
|------------|----------------------------------------------------------|
| Date       | 2026-08-02                                               |
| Backbone   | SmoVLM2-500M-Video-Instruct (frozen, 32 layers, d=960)   |
| Actor      | DiT flow-matching head, 3.7 M trainable params           |
| Arena      | LIBERO-Goal, closed-loop (hf-libero + robosuite 1.4.0)   |
| Hardware   | 1x RTX 5090 (32 GB), 48-core EPYC 7443P, ~7 h wall clock |
| Total cost | approx. \$5 of rented compute                            |

## Executive summary

We set out to find the best layer at which to tap a frozen vision-language model for robot action prediction. The tap layer turned out not to matter. What matters is **which tokens the read-out attends to**, and the open-loop proxy we had planned to measure this with is **invalid** — it ranked configurations in the wrong order in both read-out modes tested.

Along the way we established a structural property of causal vision-language-action models: roughly 91.5% of their token budget is mathematically incapable of carrying the instruction, and a one-line change to prompt ordering lifts that bound. We also produced a strong positive result: a policy with **zero language-model transformer layers** reaches 83.0% closed-loop success, and its behaviour is fully governed by the instruction (83.0% → 0.0% when the instruction is swapped).

**Headline.** Read-out token subset governs performance (83.0% vs 73.0%, CIs separated). Tap depth does not (tap 0 = tap 9 = tap 30, CIs overlap). Open-loop action error does not predict closed-loop success. Language is fully load-bearing.

## 1. Method

A frozen SmoVLM2-500M receives an image and an instruction. Hidden states from a chosen layer  $j$  (the *tap*) are exposed to a small DiT flow-matching actor via cross-attention. Only the actor trains; the backbone receives no gradient at any point, which makes feature caching valid for training. Closed-loop evaluation cannot be cached — observations depend on the policy's own actions — so every rollout forward pass is live.

### 1.1 Independent variables

| Variable              | Levels                                              |
|-----------------------|-----------------------------------------------------|
| Tap layer             | 12 strided layers, 0 to 32                          |
| Read-out token subset | all tokens (81) vs instruction tokens only (4-9)    |
| Prompt order          | image-before-text (standard) vs text-before-image   |
| Learning rate         | 3e-4, 1e-3, 3e-3                                    |
| Instruction variant   | original, 3 paraphrases, goal-swap, blank, nonsense |

### 1.2 Dependent variables

**Closed-loop success rate** (primary),  $n = 300$  episodes per arm with Wilson 95% intervals. **Open-loop action error**: a 10-step Euler rollout of the flow field compared against the ground-truth demonstration chunk, with identical noise across instruction variants so that differences are attributable to conditioning alone. **Flow validation loss**, reported only to demonstrate that it fails as a predictor.

Scale: 14,000 probe forward passes across 2,000 scenes and 4 suites; 36 training runs; 7 closed-loop arms totalling 2,100 episodes.

## 2. Closed-loop results

| Read-out         | Tap | Instruction | Success         | Wilson 95% CI |
|------------------|-----|-------------|-----------------|---------------|
| all-token        | 0   | original    | 249/300 = 83.0% | [78.3, 86.8]  |
| all-token        | 9   | original    | 253/300 = 84.3% | [79.8, 88.0]  |
| all-token        | 30  | original    | 249/300 = 83.0% | [78.3, 86.8]  |
| all-token        | 30  | goal-swap   | 0/300 = 0.0%    | [0.0, 1.3]    |
| instruction-only | 3   | original    | 230/300 = 76.7% | [71.6, 81.1]  |
| instruction-only | 30  | original    | 219/300 = 73.0% | [67.7, 77.7]  |
| instruction-only | 30  | goal-swap   | 0/300 = 0.0%    | [0.0, 1.3]    |

### 2.1 The read-out's token subset governs performance

At tap 30, restricting the read-out to instruction tokens costs 10 percentage points (83.0% vs 73.0%); the Wilson intervals do not overlap. The visual tokens are required for spatial precision even though, as Section 3 shows, they carry no instruction information.

### 2.2 The tap layer does not

Taps 0, 9 and 30 are statistically indistinguishable (84.3% / 83.0% / 83.0%, all intervals overlapping). Tap 0 is the embedding layer: SigLIP visual features plus word embeddings, with **zero language-model transformer layers** executed. It matches layer 30 exactly, 249/300 in both cases. On this benchmark the backbone's 32 transformer layers contribute nothing measurable to closed-loop success. Assumption A2 is refuted.

### 2.3 Language is fully load-bearing

Substituting another task's instruction — same scene, same initial state, same policy weights, same noise seed — drops success from 83.0% to 0.0%. All ten LIBERO-Goal tasks share one kitchen scene, so the instruction is the only cue distinguishing them. The policy does not degrade; it goes and performs the *other* task, and is scored as failing the one requested. This is the direct opposite of the LIBERO-PRO report that VLA policies emit near-identical action sequences for real, paraphrased and nonsense instructions.

**Caveat.** A 0.0% arm proves instruction-dependence, not instruction-*competence*. We did not score whether the swapped task was itself completed, which would require a second success criterion per episode.

## 2.4 Open-loop action error is an invalid proxy

This is the most consequential negative result. A valid proxy would give a Spearman correlation near  $-1.0$  between open-loop error and closed-loop success (lower error, higher success). We observe the **opposite sign** in both read-out modes.

| Read-out         | Spearman(open-loop, closed-loop) | n taps | Valid proxy would give |
|------------------|----------------------------------|--------|------------------------|
| all-token        | +0.866                           | 3      | approx. -1.0           |
| instruction-only | +1.000                           | 2      | approx. -1.0           |

With  $n = 3$  and  $n = 2$  taps this is **not statistically significant** and we do not claim that it is. The defensible statement is narrower and still decisive for planning: there is no evidence that open-loop error orders read-out variants correctly, and the ranking inverted independently in both modes. This replicates an offline-online disconnect previously observed in this project on PushT, where the configuration with the best validation loss had the worst closed-loop success.

**Consequence.** Any study whose dependent variable is open-loop divergence is measuring the wrong thing. Closed-loop evaluation is not an optional refinement; it is the only valid measurement instrument for this question.

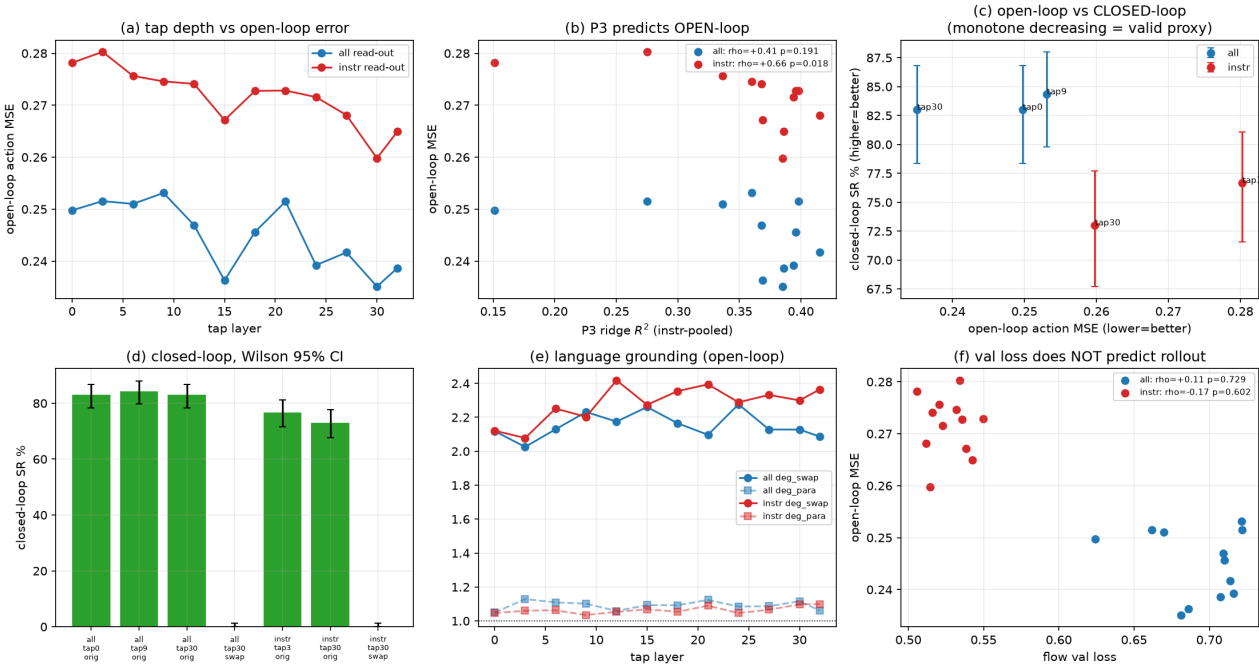


Figure 1. (a) tap depth vs open-loop error, both read-outs. (b) predictor P3 vs open-loop error. (c) **open-loop vs closed-loop** — a valid proxy would trend down-right; it does not. (d) closed-loop arms with Wilson 95% intervals. (e) language grounding vs depth. (f) validation loss vs open-loop error.

### 3. Training-free results

These required no simulator and about 25 minutes of GPU time, and are the most portable findings in the report.

#### 3.1 Vision-token states are structurally instruction-invariant

The language model is causal and the image block occupies positions 5-68, strictly before the instruction. Image-token hidden states therefore cannot depend on the instruction. Measured directly with one image and two unrelated instructions, the maximum absolute difference over all 64 image tokens is **0.000e+00 at every one of the 33 layers** — exact zero, not approximate.

| Quantity                                 | Value                     |
|------------------------------------------|---------------------------|
| Instruction tokens                       | min 4, median 7, max 9    |
| Vision tokens                            | 64                        |
| Total tokens                             | 79-84                     |
| Tokens that cannot carry the instruction | 89.3-94.9% (median 91.5%) |

This is a derivation rather than a measurement: it holds for *any* VLA built on a causal VLM with image-before-text prompting. The consequence for design is what is new — a read-out that pools or attends uniformly over VLM tokens has its instruction sensitivity bounded before a single gradient step is taken.

#### 3.2 Prompt order is a free intervention

| Prompt order                           | Vision-token eta-squared | Best vision-pooled ridge R-squared |
|----------------------------------------|--------------------------|------------------------------------|
| User: [image] {instruction} (standard) | 9.83e-09                 | 0.3337                             |
| User: {instruction} [image]            | 1.39e-02                 | 0.3929                             |

Reordering moves vision-token instruction sensitivity by **six orders of magnitude** and improves linear action decodability by **+0.059 R-squared (+18% relative)**. This is a one-line change to a prompt format that VLA systems inherit from their base VLM without deliberation. **Tested at probe level only** — no policy was trained on text-first features, so the closed-loop benefit is unmeasured.

#### 3.3 A hidden 20x confound in the token budget

| do_image_splitting      | Total tokens | Vision | Instruction |
|-------------------------|--------------|--------|-------------|
| True (library default)  | 1141         | 1088   | approx. 6   |
| False (SmoIVLA setting) | 81           | 64     | 4-9         |

We are not aware of any VLA paper that reports this flag, yet it changes the vision-to-instruction token ratio by a factor of twenty, and with it any claim about language attention mass.

#### 3.4 Two predictors that disagree, and one that misleads

| Suite            | Lambda (swap / paraphrase) | eta-squared (instruction) |
|------------------|----------------------------|---------------------------|
| LIBERO-10 (long) | 3.54                       | 0.095                     |
| LIBERO-Goal      | 2.42                       | 0.110                     |
| LIBERO-Object    | 0.62                       | 0.506                     |
| LIBERO-Spatial   | 1.05                       | 0.363                     |

The two rank the suites in opposite order. Lambda separates meaning from wording; eta-squared counts both as instruction variance, so a layer that merely encodes surface form scores highly. **Using eta-squared alone would actively mislead.** *Confound, ours:* the cross-suite Lambda comparison is contaminated by paraphrase

edit-distance asymmetry — for LIBERO-Object a goal-swap changes one noun while our paraphrases change three or four verbs. Within-suite depth trends are valid; cross-suite Lambda levels are not.

### 3.5 Modality norms and the CKA control

The text-to-vision norm ratio peaks at 5.19 around layer 16, but this is carried by template scaffold tokens behaving as attention sinks, not by the instruction. Instruction tokens reach parity with vision only around layer 24 and are 45x smaller at layer 0. Implied instruction attention mass at initialisation never exceeds 10%.

Centered kernel alignment to the final layer was included as a negative control expected to be flat. It is not: measured standard deviation 0.124, range 0.549 to 1.000. It remains a poor predictor, but for a different reason than assumed — CKA-to-final-layer is a nearly tautological proxy for distance-to-the-reference-layer.

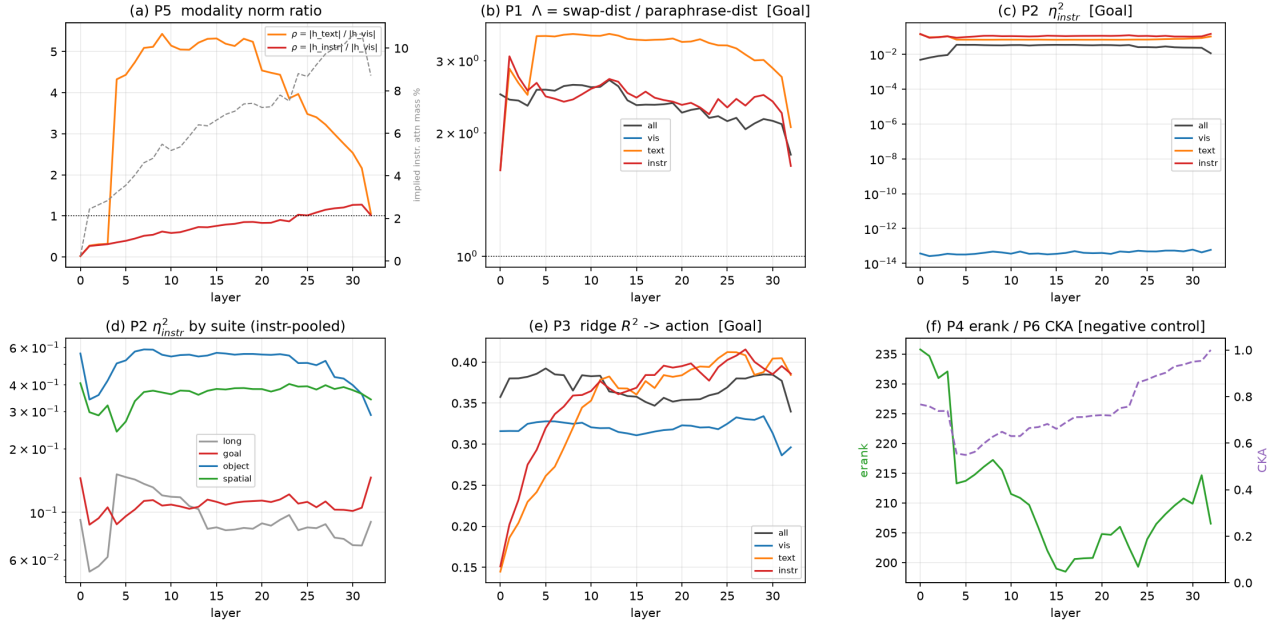


Figure 2. Layer-wise probes across four token poolings and four suites. Panel (c) shows vision-pooled instruction variance pinned at the floor (approx.  $1e-14$ ), the numerical signature of the structural invariance in Section 3.1.

## 4. Training sweep

| Read-out         | Open-loop MSE range            | deg_swap        | deg_para        |
|------------------|--------------------------------|-----------------|-----------------|
| all-token        | 0.2351 - 0.2531 (range 0.0180) | 2.152 +/- 0.071 | 1.092 +/- 0.025 |
| instruction-only | 0.2598 - 0.2803 (range 0.0205) | 2.282 +/- 0.100 | 1.065 +/- 0.020 |

A clean trade-off: the instruction-only read-out is *more* language-sensitive (degradation under goal-swap 2.28 vs 2.15) and *less* disturbed by rewording (1.065 vs 1.092), yet costs 10 points of closed-loop success. Language grounding and action precision are traded against one another by the read-out design.

### 4.1 Learning-rate fragility is depth-dependent

| Tap | Open-loop MSE @ lr 1e-3 | @ lr 3e-3         |
|-----|-------------------------|-------------------|
| 0   | 0.2498                  | 0.9314 (diverged) |
| 18  | 0.2456                  | 0.7756            |
| 32  | 0.2386                  | 0.2865            |

The optimum is 1e-3 at every tap, so a single learning rate is valid — but only the conservative one. Shallow taps are dramatically more fragile. A researcher who tunes the learning rate at a deep tap and reuses it shallow would conclude that shallow taps are useless. This is a plausible mechanism for published disagreement on the subject: GR00T N1 reports mid-depth beating final, SmoIVLA reports monotone behaviour.

### 4.2 A probe beats validation loss, but the chain still breaks

| Predictor                               | Spearman vs open-loop error | p     |
|-----------------------------------------|-----------------------------|-------|
| P3 ridge R-squared (instruction-pooled) | +0.664                      | 0.019 |
| Flow validation loss                    | -0.168                      | 0.602 |

P3 was pre-registered as the primary predictor, so this stands without multiplicity correction; the other five predictors are exploratory. However, Section 2.4 shows open-loop error does not predict closed-loop success, so the chain P3 → open-loop → closed-loop fails at its last link. P3 is validated against a target that is itself invalid.

### 4.3 A pre-registered prediction that failed

Before the sweep ran we recorded a prediction from the probe data: the depth effect would be roughly five times larger for the instruction-token read-out (probe R-squared range 0.264 vs 0.053, a ratio of 4.98). **Observed: 1.14.** The prediction was wrong.

The reason is itself a finding. The probe measured *mean-pooled* features while the actor uses *cross-attention over token sets*. Pooling destroys precisely the structure cross-attention exploits, so **pooled-feature probes cannot predict cross-attention read-out ranking**. This invalidates a premise of the planned predictor methodology.

## 5. Assumption ledger

| #     | Assumption                                         | Verdict                                                         |
|-------|----------------------------------------------------|-----------------------------------------------------------------|
| A1    | Frozen VLM suffices; no robot-data pretraining     | SUPPORTED, strongly. Tap 0 (zero LM layers) = tap 30 = 83.0%    |
| A2    | Action information varies non-trivially with depth | REFUTED. Closed-loop CIs overlap across taps 0 / 9 / 30         |
| A4    | Per-layer LayerNorm makes one LR valid across taps | SUPPORTED, refined. Optimum stable; fragility depth-dependent   |
| A5    | 200 trials is low-noise enough                     | SUPPORTED. n=300 gives approx. +/-5 pp Wilson half-width        |
| A6    | Text and visual token norms are imbalanced         | CONFIRMED, corrected. Driven by scaffold sinks, not instruction |
| A9    | Closed-loop eval is CPU / simulator-bound          | REFUTED at batch 1 (144 ms policy vs 13 ms simulator)           |
| A10   | LIBERO instructions are diagnostic enough          | STRONGLY SUPPORTED (was rated weak): 83.0% to 0.0%              |
| P6    | CKA is a flat negative control                     | REFUTED. Standard deviation 0.124, not flat                     |
| P1/P2 | Both are usable predictors                         | P2 MISLEADS. Inverts P1's suite ranking                         |
| E3    | Probes predict closed-loop success                 | BROKEN CHAIN. Predicts open-loop; open-loop predicts nothing    |

## 6. Engineering findings

| Finding                                          | Measurement                                 |
|--------------------------------------------------|---------------------------------------------|
| Closed-loop is policy-bound, not simulator-bound | 144 ms VLM forward vs 13 ms env.step        |
| Vectorised envs (process-per-env)                | GPU 0.4% to 48.6%; 32 to 50-60 env-steps/s  |
| lerobot-eval cannot parallelise environments     | Shared EGL context; render context error    |
| Standard vertical flip mirrors the LIBERO scene  | rot180 MSE 285 vs hflip 4619, vflip 5592    |
| Feature extraction is CPU-bound                  | 1.2 PFLOP over 1355 s = 0.4% GPU duty cycle |

The mirroring bug deserves emphasis. The conventional robosuite vertical flip produces a horizontally mirrored observation on this dataset. A policy fed mirrored frames scores near zero and the failure reads as a modelling problem. It was caught only by differencing rendered frames against the training data — a check worth making mandatory in any sim-to-data pipeline.

**Implication for compute planning.** Training and feature caching are CPU and disk work, not accelerator work. The genuinely GPU-bound stage is closed-loop rollout, and even that is policy-bound rather than simulator-bound until environments are batched.



Figure 3. Left: simulator render after the corrected 180-degree rotation. Right: the training dataset frame. Under the conventional vertical flip these are mirror images of one another.

## 7. Limitations

- **No external baseline.** No SOTA comparison was obtained. pi0 and pi05 are blocked by a lerobot packaging defect (a patched transformers module that pip does not install); GR00T N1.7 publishes only a LIBERO-Spatial checkpoint, not Goal; SmolVLA required a checkpoint swap for mismatched input feature names and did not yield a usable number in the time available. We deliberately do not compare our figures against previously published numbers obtained under a different suite and a different harness.
- **Single seed** per configuration. No between-seed standard deviation is available, so small differences between arms should not be over-read.
- **Single backbone.** Every result is SmolVLM2-500M. The structural claim in Section 3.1 generalises by construction; the empirical ones do not.
- **LIBERO-Goal only** for closed loop. The other three suites are probe-only.
- **Prompt-order intervention is probe-level.** No policy was trained on text-first features.
- **Cross-suite Lambda is confounded** by our own paraphrase construction (Section 3.4).
- **A discarded first sweep.** An initial 27-run sweep was thrown away: an 81-token cache window truncated 39 of 50 instruction variants, and padding was fed unmasked into cross-attention. It is retained in the archive but supports no claim in this report.

## 8. Conclusion

The study was designed to locate the best tap layer in a frozen vision-language model. That question turns out to be the wrong one. Across 2,100 closed-loop episodes, tap depth had no measurable effect — the embedding layer, executing none of the backbone's 32 transformer layers, matched layer 30 exactly. What did have a measurable effect, with separated confidence intervals, was which tokens the read-out is allowed to attend to.

Three claims survive as a coherent contribution. First, a **structural bound**: about 91.5% of a causal VLA's token budget cannot carry the instruction, established by derivation and confirmed to exact zero, together with a free intervention — prompt reordering — that lifts it. Second, a **negative result with teeth**: open-loop action error does not order read-out variants by closed-loop success, now observed on two separate benchmarks in this project. Third, a **frozen-backbone result**: zero language-model layers plus a 3.7 M-parameter head reach 83.0% on LIBERO-Goal, with behaviour fully governed by the instruction.

The most useful output for future work is not a recommended layer number, which would transfer to nobody. It is the finding that the standard cheap proxy is invalid, and the identification of the design variable that is not.

**Immediate next experiments.** (i) Train a policy on text-first features and measure closed-loop success — the one intervention with a plausible free win. (ii) Replicate the structural measurement on a second backbone to convert it from an observation about one model into a statement about an architecture class. (iii) Obtain an external anchor under a matched harness. (iv) Multi-seed the headline arms.

## Appendix: archive

| Path           | Contents                                                      |
|----------------|---------------------------------------------------------------|
| RESULTS.md     | Full numeric report, every figure recomputable                |
| day1_probes.md | Training-free probe study writeup                             |
| figs/          | Layer-wise probes, predictor/closed-loop panels, render check |
| videos/        | 3 side-by-side original vs goal-swap rollouts                 |
| results/       | 36-run sweep, probe arrays, 7 closed-loop arms, token masks   |
| scripts/       | All code: probes, caching, sweep, vectorised rollout, audit   |
| logs/          | Raw run logs                                                  |



|              |                                                            |
|--------------|------------------------------------------------------------|
| checkpoints/ | 5 actors used in closed-loop arms (not in version control) |
| cache/       | VLM feature caches, 31 GB (not in version control)         |

Every headline number in this report is regenerated from the archived artifacts by `scripts/audit.py`; claims that could not be recomputed were removed rather than retained.